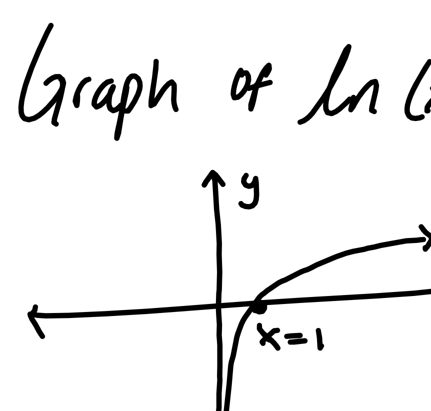
Derivatives of $\ln(x)$

Section 5.5

11/11/2025

Graph of $\ln(x)$



Properties of $\ln x$

1) $\ln(1) = 0$
2) $\ln(e^p) = p$
3) $\ln(ab) = \ln(a) + \ln(b)$
4) $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$
5) $\ln(a^b) = b\ln(a)$

$\ln \frac{(x^2+3)^2}{\sqrt[3]{x^2+1}} =$

$\ln(x^2+3)^2 - \ln(\sqrt[3]{x^2+1}) =$

$2\ln(x^2+3) - \frac{1}{3}\ln(x^2+1)$

$2\ln(x+3) + \frac{1}{2}\ln(x-2) =$

$\ln((x+3)^2(\sqrt{x-2}))$

$\ln(x+1) - 3 = 0$

$\ln(x+1) = 3$

$e^3 = x+1$

$$\frac{d}{dx}\ln x = \frac{1}{x}$$

$x > 0$

$$\frac{d}{dx}\ln u = \frac{u'}{u}$$

$f(x) = \ln(2x)$

$f'(x) = \frac{2}{2x} = \frac{1}{x}$

$y = \ln(x^2+1) =$

$y' = \frac{2x}{x^2+1}$

$f(x) = x\ln x$

$f'(x) = \ln x + 1$

$f(x) = \ln(\sqrt{x+1}) = \frac{1}{2}(\ln(x+1))$

$f'(x) = \frac{1}{2(x+1)}$

$y = \ln(\ln x) =$

$y' = \frac{\frac{1}{x}}{\ln(x)} = \frac{1}{x\ln(x)}$

$y = (\ln x)^3 =$

$y' = 3(\ln x)^2 \cdot \frac{1}{x}$

$y = \ln(x^3)$

$y' = \frac{3}{x}$

$f(x) = \ln\left(\frac{x^2\sqrt{3x-2}}{\sqrt{(x^2+1)^3}}\right)$

$= \left(2\ln(x) + \frac{1}{2}\ln(3x-2)\right) - \left(\frac{3}{2}\ln(x^2+1)\right)$

$f'(x) = \frac{2}{x} + \frac{3}{2(3x-2)} - \frac{3\cancel{2}x}{\cancel{2}(x^2+1)}$

$f(x) = \ln(\sin x) =$

$f'(x) = \frac{\cos x}{\sin x} = \frac{1}{\tan x} = \cot x$

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tangent line of

$y = \ln(x^3)$ @ $(1,0)$

$y' = \frac{3}{x}$

$y - 0 = 3(x-1)$

$x^3 + 2\ln y + y^3 = 4$

$\frac{dy}{dx}$ in terms of x & y

$3x^2 + \frac{2}{y}\cdot\frac{dy}{dx} + 3y^2\cdot\frac{dy}{dx} = 0$

$\frac{dy}{dx}\left(\frac{2}{y} + 3y^2\right) = -3x^2$

$\frac{dy}{dx} = \frac{-3x^2}{\frac{2}{y} + 3y^2} = \frac{-3x^2y}{2+3y^3}$